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import React, { useState, useRef, useEffect } from 'react';

// Artifact: Spectral Graviton Core
// Tier: 6
// Synthesized by AlchemyForge.app Engine
// -----

export default function SpectralGravitonCoreCard() {
  const cardRef = useRef(null);
  const rARef = useRef(null);
  const [tilt, setTilt] = useState({ active: false, mouseX: 150, mouseY: 210 });
  const [physicsState, setPhysicsState] = useState({ currentX: 0, currentY: 0,
targetX: 0, targetY: 0 });
  const [glareAngle, setGlareAngle] = useState(0);

  // Archangel Derived Physics Parameters
  const mass = 5;
  const friction = 0.8;
  const glareType = 'linear-sheen';
  const cursorBehavior = 'neutral';

  useEffect(() => {
    const loop = () => {
      setPhysicsState(prev => {
        const lerpFactor = Math.max(0.01, Math.min(1, 1 / (mass * friction)));
        const dx = prev.targetX - prev.currentX;
        const dy = prev.targetY - prev.currentY;
        return {
          ...prev,
          currentX: prev.currentX + dx * lerpFactor,
          currentY: prev.currentY + dy * lerpFactor
        };
      });
      rARef.current = requestAnimationFrame(loop);
    };
    rARef.current = requestAnimationFrame(loop);
    return () => cancelAnimationFrame(rARef.current);
  }, [mass, friction]);

  const handleMouseMove = (e) => {
    if (!cardRef.current) return;
    const rect = cardRef.current.getBoundingClientRect();
    const x = e.clientX - rect.left;
    const y = e.clientY - rect.top;

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const centerX = rect.width / 2;
const centerY = rect.height / 2;

const angle = Math.atan2(y - centerY, x - centerX) * (180 / Math.PI);
setGlareAngle(angle);

const maxTilt = 35;
let targetTiltX = ((y - centerY) / centerY) * -maxTilt;
let targetTiltY = ((x - centerX) / centerX) * maxTilt;

if (cursorBehavior === 'repel') { targetTiltX *= -1.5; targetTiltY *= -1.5; }
else if (cursorBehavior === 'attract') { targetTiltX *= 1.5; targetTiltY *=
1.5; }
else if (cursorBehavior === 'heavy') { targetTiltX *= 0.5; targetTiltY *= 0.5;
}

setTilt({ active: true, mouseX: x, mouseY: y });
setPhysicsState(prev => ({ ...prev, targetX: targetTiltX, targetY: targetTiltY
}));
};

const handleMouseLeave = () => {
  setTilt(prev => ({ ...prev, active: false }));
  setPhysicsState(prev => ({ ...prev, targetX: 0, targetY: 0 }));
};

const getTransitionStyle = () => {
  if (tilt.active) return 'transform 0.1s cubic-bezier(0.25, 0.8, 0.25, 1)';
  return 'transform 0.6s cubic-bezier(0.25, 0.8, 0.25, 1)';
};

const getHoverTransform = () => {
  let base = `perspective(1200px) rotateX(${physicsState.currentX}deg)
rotateY(${physicsState.currentY}deg)`;
  if (tilt.active) {
    return `${base} scale3d(1.05, 1.05, 1.05) translateZ(30px)`;
  }
  return `perspective(1200px) rotateX(0deg) rotateY(0deg) scale3d(1, 1, 1)
translateZ(0)`;
};

let activeGlareStyle = {
  position: 'absolute', inset: 0, pointerEvents: 'none', zIndex: 20,
  opacity: tilt.active ? 1 : 0, transition: 'opacity 0.4s ease-in-out',
  transform: 'translateZ(1px)'
};

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    if (glareType === 'radial-spotlight') {
        activeGlareStyle.background = `radial-gradient(circle at ${tilt.mouseX}px
        ${tilt.mouseY}px, rgba(255,255,255,0.4) 0%, transparent 60%)`;
        activeGlareStyle.mixBlendMode = 'overlay';
    } else if (glareType === 'aurora-wash') {
        activeGlareStyle.background = `linear-gradient(${glareAngle}deg, #00ffff00 0%,
        #00ffff80 50%, #7b00ff00 100%)`;
        activeGlareStyle.mixBlendMode = 'color-dodge';
    } else if (glareType === 'none') {
        activeGlareStyle.display = 'none';
    } else {
        activeGlareStyle.background = `linear-gradient(${glareAngle}deg,
        rgba(255,255,255,0) 0%, rgba(255,255,255,0.92) 50%, rgba(255,255,255,0) 100%)`;
        activeGlareStyle.mixBlendMode = 'color-dodge';
    }

    return (
        <div style={{ position: 'relative', width: '300px', height: '420px' }}>
            {/* Archangel Generated Dynamic CSS */}
            <style dangerouslySetInnerHTML={{ __html: `
                @keyframes customPulse-SpectralGravitonCore { 0%, 100% { transform:
                scale(1); } 50% { transform: scale(1.03); } }
                .custom-anim { animation: customPulse-SpectralGravitonCore 4s ease-in-out
                infinite; }
            ` }} />

            {/* Archangel Generated Dynamic SVG Filter */}
            <svg style={{ position: 'absolute', width: 0, height: 0 }}>
                <filter id="filter-SpectralGravitonCore" dangerouslySetInnerHTML={{
                __html: `
                    <feTurbulence type="fractalNoise" baseFrequency="0.01" numOctaves="3"
                    result="noise" />
                    <feDisplacementMap in="SourceGraphic" in2="noise" scale={tilt.active ? 15 :
                    5} xChannelSelector="R" yChannelSelector="G" />
                ` }} />
            </svg>

            <div
                ref={cardRef}
                onMouseMove={handleMouseMove}
                onMouseLeave={handleMouseLeave}
                className="custom-anim"
                style={{
                    position: 'relative',
                    width: '100%',
                    height: '100%',
                    borderRadius: '2rem',

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border: '2px solid #00ffff',
overflow: 'hidden',
display: 'flex',
flexDirection: 'column',
justifyContent: 'space-between',
padding: '24px',
cursor: 'pointer',
transform: getHoverTransform(),
transition: getTransitionStyle(),
background: `radial-gradient(circle at center, #000000 0%, #0a0015 30%,
#4b0082 60%, #00ffff 100%)`,
backgroundSize: 'cover',
backgroundPosition: 'center',
backgroundBlendMode: 'normal',
boxShadow: tilt.active
  ? `0 20px 60px #00ffff80, inset 0 0 20px #00ffff40`
  : `0 10px 30px rgba(0,0,0,0.5)` ,
filter: 'url(#filter-SpectralGravitonCore)',
color: 'white',
fontFamily: 'system-ui, sans-serif',
boxSizing: 'border-box'
}}

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>

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<div style={activeGlareStyle} />
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/* Content Header */
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<div style={{ position: 'relative', zIndex: 10, transform: tilt.active ?
'translateZ(40px)' : 'translateZ(0)', transition: 'transform 0.3s' }}>

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  <div style={{ display: 'flex', justifyContent: 'space-between',
alignItems: 'flex-start', marginBottom: '24px' }}>

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    <div style={{
      padding: '12px', borderRadius: '16px', background:
'rgba(0,0,0,0.6)', backdropFilter: 'blur(12px)',
      border: '1px solid rgba(255,255,255,0.1)', color: '#00ffff',
      boxShadow: '0 0 20px rgba(0,0,0,0.6)'
    }}>

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</div>
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    <span style={{
      fontSize: '10px', fontWeight: 900, letterSpacing: '0.1em',
textTransform: 'uppercase',
      padding: '6px 12px', borderRadius: '999px', background:
'rgba(0,0,0,0.8)',
      backdropFilter: 'blur(12px)', border: '1px solid
rgba(255,255,255,0.1)'
    }}>

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        </span>
    </div>

    <div style={{
        padding: '16px', borderRadius: '16px', background: 'rgba(0,0,0,0.4)',
        backdropFilter: 'blur(12px)',
        border: '1px solid rgba(255,255,255,0.05)'
    }}>
        <h3 style={{ fontSize: '24px', fontWeight: 900, margin: '0 0 4px 0',
        textShadow: '0 0 20px #00ffff80' }}>
            Spectral Graviton Core
        </h3>
        <p style={{ fontSize: '12px', fontWeight: 700, textTransform:
        'uppercase', letterSpacing: '0.05em', color: '#ffffff', margin: 0 }}>
            Where Fourier-transformed luminosity meets the crushing focus of an
            event horizon.
        </p>
    </div>
</div>

    { /* Content Body */ }

    <div style={{ position: 'relative', zIndex: 10, marginTop: 'auto',
    transform: tilt.active ? 'translateZ(50px)' : 'translateZ(0)', transition:
    'transform 0.3s' }}>
        <div style={{
            padding: '16px', borderRadius: '16px', background: 'rgba(0,0,0,0.4)',
            backdropFilter: 'blur(12px)',
            border: '1px solid rgba(255,255,255,0.05)'
        }}>
            <p style={{ fontSize: '13px', lineHeight: 1.6, fontWeight: 500,
            margin: 0, color: '#e2e8f0' }}>
                A Tier 6 synthesized node that manifests as a refractive vortex. It
                utilizes Fourier-series wave analysis to modulate light intensity as it approaches
                the central singularity. Upon user interaction, the component executes a 'Photonic
                Spaghettification' sequence, warping the surrounding UI grid into iridescent
                kinetic fluid streams that spiral into a high-density dark-matter core before
                oscillating back into prismatic clarity.
            </p>
        </div>
    </div>
</div>
</div>
);
}

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